Personal Statement — MSc in Computer Science, City University of Hong Kong

In the smart gardening system I built on OpenHarmony, a YOLOv5 detector running on an RK3588 board decides when a pump runs. Making that loop trustworthy turned out to depend less on the model than on everything around it: the confidence and timestamp attached to each observation, the rule-based fast path that overrides the agent in extreme states, and the state machine on the Hi3861 that keeps the pump safe when the decision layer is uncertain. Every robot I have worked on since has repeated the pattern: reliable physical behaviour depends on the computer systems connecting perception, algorithms and actuators, on dependable interfaces, consistent state representations and execution paths that remain safe when conditions are abnormal. I am applying to the MSc in Computer Science at City University of Hong Kong to strengthen my foundations in software engineering, data engineering and artificial intelligence, and to apply them to robotics algorithms and intelligent systems.

I am a final-year Computer Science and Technology student at Shenzhen Technology University, ranked 7th out of 156 students in my major. Alongside the core CS curriculum, I took Principles of Automatic Control and Modern Control Theory. During my internship at the Shenzhen Institute of Artificial Intelligence and Robotics for Society, I was responsible for chassis balance control on a three-omni-wheel ballbot carrying a dual-arm upper body. I built the control state from IMU and wheel feedback, designed and deployed a position–attitude cascaded PID controller, and allocated the control actions through the three-wheel kinematics. I also modelled the chassis as a four-state inverted pendulum and used LQR to analyse the relationship between controller gains, stability and actuator output, while estimating arm-induced centre-of-mass changes from the URDF model with Pinocchio and feeding them forward as gravity compensation.

My current internship with the Kaihong M-Robots team has further shaped my understanding of robot software systems. I contribute to an OpenHarmony platform for heterogeneous robots, working on device-capability abstraction, the mapping of ROS/ROS2 motion commands to low-level drivers, and state synchronisation across sensors and actuators. I have also worked with device discovery, capability registration and task dispatch over M-DDS. These tasks involve more than connecting interfaces: they require reusable software boundaries across different hardware, time scales and operating states, while ensuring that task plans pass checks before entering physical execution. They have made clear to me that, as a robot platform scales, networking, data consistency, security policy and software-engineering practice become as important as the control algorithm itself.

My other projects reinforce this perspective from the algorithmic side. On a car-like mobile robot, I built an autonomous-navigation chain, from EKF/AMCL localisation through Hybrid-A* global planning and TEB local trajectory optimisation to MPC tracking control, so that the vehicle could locate itself, plan around obstacles and follow the planned path on real hardware. In a dual-arm robot project, I solved constrained inverse kinematics in MuJoCo/Mink so that VR controller motion drives the end-effectors without velocity spikes near singularities, and built a gravity-compensated leader–follower controller for hand-guided teaching. Together with the gardening system, these experiences allow me to view robotics through control, planning, perception and platform software, while also showing me that I need systematic training in machine-learning foundations, data processing and scalable software design.

The MSc in Computer Science programme is built around computer-science foundations: the core courses Software Engineering, Data Engineering and Computer Networks and Internets, elective streams in artificial intelligence, data science and information security, and Project or Research/Internship Project courses. Software Engineering and Data Engineering are where I want to strengthen my methods for designing and validating larger software systems, and the Artificial Intelligence Stream, starting from Machine Learning: Principles and Practice, is where I want to replace practical familiarity with a principled understanding of machine learning. I would then use the Project course for a robotics project in perception, navigation or platform software. Compared with a programme tied to one particular robot mechanism, its computer-science training is better suited to building algorithms and software foundations that transfer across robot platforms.

After graduation, I intend to work in the Greater Bay Area on robotics algorithms and intelligent systems, focusing on perception, navigation and motion control for mobile robots, and to grow into an engineer who can connect algorithms, software platforms and physical execution. My preparation combines control, navigation and robot-platform work on real hardware. From the MSc in Computer Science programme, I seek the systematic and scalable foundation needed for the next stage of my work in robotics algorithms.